



TRS-80 CoCo 1 / 2 / 3

COLOR BASIC · EXTENDED · DISK EXTENDED · SUPER EXTENDED (COCO 3) · 6809E

Tags: [E] Extended Color BASIC · [D] Disk ECB · [3] CoCo 3 Super ECB. Base Color BASIC runs on every model; ECB adds hi-res PMODE graphics; the CoCo 3 adds 16-colour HSCREEN modes and 40/80-column text. = REM.

PROGRAM CONTROL

RUN	Start; RUN 100 from a line
LIST	LIST 10-50 range
NEW	Erase program
CLEAR n[,top]	Reserve string space / set RAM top
CONT	Resume after STOP / BREAK
STOP / END	Halt program
DEL a-b	Delete line range
EDIT n	Line editor (C D I L X K)
RENUM [E]	Renummer lines
TRON / TROFF [E]	Trace on / off
EXEC addr	Run machine code at addr

[E]=Extended Color BASIC · [D]=Disk ECB · [3]=CoCo 3 Super ECB.

BASIC STATEMENTS

PRINT ?	; joins, , zones
PRINT@ p,	Print at screen pos 0-511
INPUT	Prompt & read
LINE INPUT [E]	Read a whole line incl. commas
LET	Assign (optional)
REM '	Comment
GOTO GOSUB	Jump / call; RETURN
ON x GOTO	/ GOSUB a,b,c
FOR...NEXT	FOR I=1 TO 9 STEP 2
IF...THEN[...ELSE]	ELSE is [E]
DATA READ	RESTORE rewinds
DIM	DIM A(20),N\$(5)
POKE a,v	Write byte

FUNCTIONS

ABS SGN INT	magnitude/sign/floor
SQR SIN COS [E]	root, trig (TAN ATN EXP LOG [E])
RND(n)	n>1 int 1..n; 0 = 0..1
PEEK(a) MEM	read byte / free bytes
LEN LEFT\$ RIGHT\$ MID\$	MID\$(A\$,3,2)
STR\$ VAL ASC CHR\$	convert num/char
INKEY\$	Scan keyboard (no wait)
JOYSTK(n)	Joystick axis 0-3 → 0-63
STRING\$ HEX\$ [E]	repeat char / hex
VARPTR TIMER [E]	var address / jiffy timer
USR(x)	Call ML routine

TEXT & SEMIGRAPHS

32×16

CLS c	Clear to colour 0-8
PRINT@ p	Text position 0-511
SET(x,y,c)	Light cell: x 0-63, y 0-31
RESET(x,y)	Clear a cell
POINT(x,y)	Colour at cell (-1 if text)
SOUND t,d	Tone 1-255, duration 1-255
PLAY "... " [E]	Music string (see DRAW/PLAY)
AUDIO/MOTOR ON	Cassette monitor / motor

VDG COLOUR SET

CLS / SET 0-8

0 Black	1 Green
2 Yellow	3 Blue
4 Red	5 Buff
6 Cyan	7 Magenta
8 Orange	

Same nine 6847 colours used by CLS, SET and the hi-res COLOR command.

HI-RES GRAPHICS

[E]

PMODE m,page	Mode 0-4, start page 1-8
PCLEAR n	Reserve n graphics pages
PCLS [c]	Clear hi-res screen
SCREEN t,cs	t 0=text/1=gfx; cs colour-set 0/1
COLOR f,b	Foreground / background
LINE(x,y)-(x,y),PSET	Add ,B box / ,BF filled
CIRCLE(x,y),r,c,hw	+ start,end arc fractions
PAINT(x,y),c,b	Flood fill to border b
PPOINT(x,y)	Read pixel colour
GET / PUT	Save / stamp a rectangle
PCOPY s TO d	Copy graphics page

PMODE RESOLUTIONS

[E]

PMODE 0	128×96 · 2 colour · 1 page
PMODE 1	128×96 · 4 colour · 2 pages
PMODE 2	128×192 · 2 colour · 2 pages
PMODE 3	128×192 · 4 colour · 4 pages
PMODE 4	256×192 · 2 colour · 4 pages

All addressed as 256×192. Each page = 1536 bytes; reserve with PCLEAR.

DRAW & PLAY STRINGS

[E]

DRAW: U D L R n	Up/Down/Left/Right n dots
E F G H n	Diagonals (NE/SE/SW/NW)
M x,y	Move to point (± = relative)
C n A n S n	Colour / Angle 0-3 / Scale
B N	Blank move / No position update
PLAY: A-G	Notes (# + sharp, - flat)
O n L n	Octave 1-5 / note Length
T n V n P n	Tempo / Volume 1-31 / Pause

DISK EXTENDED BASIC

[D]

DSKINI n	Format (initialise) disk in drive n
DIR n	List directory of drive n
LOAD f,R	Load BASIC (R = run)
SAVE f,A	Save (A = ASCII)
RUN f,R	Load & run a program
LOADM f / SAVEM f,a1,a2,x	ML load / save range+exec
KILL f	Delete file
RENAME f1 TO f2	Rename
COPY f1 TO f2	Duplicate file
BACKUP n TO m	Copy whole disk
VERIFY ON/OFF DRIVE n	Write-check / default drive
OPEN/CLOSE/PRINT#/INPUT#	Sequential file I/O (#-1..)

Also MERGE, FILES, LINE INPUT#, EOF, DSKI\$/DSK0\$ (direct sector I/O).

COCO 3 SUPER ECB

[3]

HSCREEN m	Hi-res mode 0-4 (see table)
HCLS [c] HCOLOR f,b	Clear / set colours (slots 0-15)
HLINE(x,y)-(x,y),PSET	+ ,B / ,BF
HCIRCLE / HPAINT	Circle / flood fill
HDRAW "... "	Hi-res DRAW string
HGET / HPUT / HBUFF	Block copy via buffers
HPRINT(c,r),str\$	Text on hi-res screen
LOCATE c,r ATTR f,b	Text cursor / colour+blink+ul
PALETTE s,c	Slot 0-15 → colour 0-63
WIDTH 32/40/80	Text columns
ON BRK / ON ERR GOTO	Trap BREAK / errors

RGB and CMP set the palette-numbering for RGB vs composite monitors.

COCO 3 HSCREEN MODES

[3]

HSCREEN 0	Text mode
HSCREEN 1	320×192 · 4 colour
HSCREEN 2	320×192 · 16 colour
HSCREEN 3	640×192 · 2 colour
HSCREEN 4	640×192 · 4 colour

16 active colours chosen from 64 via PALETTE.

MEMORY & SPEED

POKE

1024 (\$400)	Text screen, 32×16 (512 bytes)
POKE 65495,0	CoCo 1/2 high speed (~1.8 MHz)
POKE 65494,0	CoCo 1/2 back to normal
POKE 65497,0	CoCo 3 double speed (stable)
POKE 65496,0	CoCo 3 back to 0.89 MHz
RESET button	Warm restart; hold for cold on CoCo 3

On CoCo 1/2 the 65495 trick can disturb cassette/disk I/O — switch back before loading.

QUICK-FACTS

CoCo 1 (1980)	6809E 0.89 MHz · 4-64K · Color/ECB
CoCo 2 (1983)	Same, smaller & cheaper · ECB std
CoCo 3 (1986)	6809E · 128/512K · GIME · RGB · 40/80-col
Video	MC6847 VDG (CoCo 3: GIME) · SAM MC6883
Disk	RS-DOS (Disk ECB), drives 0-3

EDITING & KEYS

BREAK	Stop a running program
SHIFT + @	Pause / resume output
← → ↑ ↓	Cursor / edit movement
SHIFT + 0	Toggle UPPER / lower case
CLEAR	Clear screen (key)
CTRL (CoCo 3)	Extra characters / attributes